SIMULATION PROTOCOL: RESONANCE-CORE

Computational Stress-Test for the Law of Dependent Resonance

1. System Initialization (The "Stage")

- **Ensemble Type:** NPT (Constant Number of particles, Pressure, and Temperature).
- **Initial Temp:** $0.1\ K$ (Near Absolute Zero to minimize thermal noise).
- **Time Step:** *0.5 femtoseconds* (10^{-15} *s*) to capture high-frequency magnetic flips.
- **Periodic Boundary Conditions (PBC):** Enabled in all three dimensions (x, y, z).

2. Atomic Topology (The "Dodecahedron")

Construct the lattice structure with the following parameters:

- **Unit Cell:** Dodecahedral.
- **Dopant Placement:** Central *FeCo* nucleus at (0, 0, 0) of each cell.
- **Lattice Constant:** $a = 0.5\ nm$.
- **Rotation Instruction:** Rotate every second cell layer by *31.7°* on the z-axis and *15%* volumetric offset on the x-axis.

3. Potential Energy Definitions (The "Rules")

Define atomic interactions using these specific physics parameters:

- **Lennard-Jones Potential:** For the Quartz matrix *(SiO_2)* structural repulsion.
- **Dipolar Interaction:** Define ***FeCo*** nuclei with a magnetic moment of ***μ = 2.2 Bohr magnetons***.
- **The "Frustration" Force:** Apply a Hamaker Constant for overlapping fields. Calculate Magnetic Torque *($T = m \times B$)* between tilted dodecahedrons.

4. The "Spark" Protocol (The Compression)

The core activation sequence:

- **Ramp Pressure:** Increase external pressure from ***0 GPa*** to ***15 GPa*** over 100,000 time steps.
- **Trigger Event:** Monitor for the "Piezoelectric Snap" at ***12 GPa***; ***SiO_2*** atoms must shift by ***2.4%***.
- **Success Metric:** Sustained self-oscillation of nuclei after threshold is reached without external input.

5. Data Extraction (The "Proof")

Required output visualizations:

- **Total Energy vs. Time:** Tracking energy harvest from internal stress.
- **Radial Distribution Function (RDF):** Proof of structural integrity of the "Tilted Dodecahedron" grid.

- **Magnetic Flux Density Map:** Visualization of "Magnetic Braiding" between atoms.